THE KOSZUL LOCUS OF A MAP OVER A FLAT FINITELY PRESENTED BASE FAMILY IS OPEN AND STABLE UNDER BASE CHANGE (SPACES)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11382-11398 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11400-11438, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete); schemes twin is imported as the load-bearing input, so its own label is the uncounted side of this pair per ruling v3.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Morphisms, lemma base-change-lci-fibres (schemes twin); chapter definition-lci; Properties of Spaces: open-subspaces, etale-open.

Source reference. Original lemma 069F: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11153–11164. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

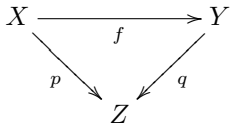
Source reference. Original lemma 06B8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11438–11438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03IR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11438–11438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11438–11438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram*



of algebraic spaces over S. Assume that both p and q are flat and locally of finite presentation. Then there exists an open subspace $U(f) \subset X$ such that $|U(f)| \subset |X|$ is the set of points where f is Koszul. Moreover, for any morphism of algebraic spaces $Z' \to Z$, if $f' : X' \to Y'$ is the base change of f by $Z' \to Z$, then $U(f')$ is the inverse image of $U(f)$ under the projection $X' \to X$.

Proof. This lemma is the analogue of More on Morphisms, Lemma 06B8 and in fact we will deduce the lemma from it. By Definition 069F the set $\{x \in |X| : f \text{ is Koszul at } x\}$ is open in $|X|$ hence by Properties of Spaces, Lemma 03BZ it corresponds to an open subspace $U(f)$ of X. Hence we only need to prove the final statement.

Choose a scheme W and a surjective étale morphism $W \to Z$. Choose a scheme V and a surjective étale morphism $V \to W \times_Z Y$. Choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. Finally, choose a scheme W' and a surjective étale morphism $W' \to W \times_Z Z'$. Set $V' = W' \times_W V$ and $U' = W' \times_W U$, so that we obtain surjective étale morphisms $V' \to Y'$ and $U' \to X'$. We will use without further mention an étale morphism of algebraic spaces induces an open map of associated topological spaces (see Properties of Spaces, Lemma 03IR). Note that by definition $U(f)$ is the image in $|X|$ of the set T of points in U where the morphism of schemes $U \to V$ is Koszul. Similarly, $U(f')$ is the image in $|X'|$ of the set T' of points in U' where the morphism of schemes $U' \to V'$ is Koszul. Now, by construction the diagram

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ V' & \longrightarrow & V \end{array}$$

is cartesian (in the category of schemes). Hence the aforementioned More on Morphisms, Lemma 06B8 applies to show that T' is the inverse image of T. Since $|U'| \to |X'|$ is surjective this implies the lemma. □